

run-llama / llama_index

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feat(callbacks): add TokenBudgetHandler for cost governance #20546

Merged logan-markewich merged 2 commits into run-llama:main from Jayachander123:feat/token-budget-handler 7 minutes ago

Conversation 1 Commits 2 Checks 14 Files changed 2 +92 -0

Jayachander123 commented 3 days ago

Contributor

Description

This PR introduces a new `TokenBudgetHandler` to the `llama-index-core` callbacks.

Motivation:

Currently, `TokenCountingHandler` provides visibility into token usage but lacks enforcement mechanisms. In production environments and automated pipelines, this can lead to runaway costs if an agent enters an infinite loop or processes unexpectedly large documents.

Changes:

- Implemented `TokenBudgetHandler`, a subclass of `TokenCountingHandler`.
- It monitors usage in real-time via the `on_event_start` hook.
- It raises a `ValueError` immediately when the budget is exceeded, preventing further API calls.

New Package?

Did I fill in the `tool.llamahub` section in the `pyproject.toml` and provide a detailed `README.md` for my new integration or package?

☐ Yes☒ No

Version Bump?

Did I bump the version in the `pyproject.toml` file of the package I am updating? (Except for the `llama-index-core` package)

☐ Yes☒ No

Type of Change

Please delete options that are not relevant.

☐ Bug fix (non-breaking change which fixes an issue)☒ New feature (non-breaking change which adds functionality)☐ Breaking change (fix or feature that would cause existing functionality to not work as expected)☐ This change requires a documentation update

How Has This Been Tested?

Your pull-request will likely not be merged unless it is covered by some form of impactful unit testing.

☒ I added new unit tests to cover this change☐ I believe this change is already covered by existing unit tests

Test Details:

- Created `tests/callbacks/test_token_budget.py`.
- Verified that `ValueError` is raised exactly when the token count exceeds the budget.
- Verified that standard `TokenCountingHandler` behavior remains unchanged.

Suggested Checklist:

☒ I have performed a self-review of my own code☒ I have commented my code, particularly in hard-to-understand areas☐ I have made corresponding changes to the documentation☐ I have added Google Colab support for the newly added notebooks.☒ My changes generate no new warnings☒ I have added tests that prove my fix is effective or that my feature works☒ New and existing unit tests pass locally with my changes☒ I ran `uv run make format`; `uv run make lint` to appease the lint gods

Reviews

logan-markewich ✓AstraBert ●

Assignees

No one assigned

Labels

lgtnsize:M

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

Notifications

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3 participants

feat(callbacks): add TokenBudgetHandler for cost governance 448cac7

dosubot bot added the size:M label 3 days ago

AstraBert approved these changes yesterday View reviewed changes

dosubot bot added the lgtn label yesterday

logan-markewich reviewed yesterday View reviewed changes

llama-index-core/llama_index/core/callbacks/token_counting.py Outdated

291 + verbose: bool = False,292 +) -> None:293 + super().__init__(tokenizer=tokenizer, verbose=verbose)

294

+

self.token_budget = token_budget

 **logan-markewich** yesterday

Collaborator

...

I would rather just add a token budget param (optional) to the existing token counting handler

Note that this is also using the older (deprecated) callback system.

Users who want to count tokens in the newer instrumentation system are guided to just implement it

<https://colab.research.google.com/drive/1QV01kCEncYZ0Ym6o6reHPcfizSVxsQg?usp=sharing>





Reply...

 feat(callbacks): add token budget enforcement to TokenCountingHandler ✓ 75cd660

 Jayachander123 requested a review from AstraBert 18 hours ago

  **logan-markewich** approved these changes 40 minutes ago View reviewed changes

  **logan-markewich** merged commit e449faf into run-llama:main 7 minutes ago

11 checks passed

View details

Pull request successfully merged and closed

You're all set — the branch has been merged.



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Comment

Remember, contributions to this repository should follow its [contributing guidelines](#), [security policy](#), and [code of conduct](#).

 **ProTip!** Add [.patch](#) or [.diff](#) to the end of URLs for Git's plaintext views.